

Arya Honraopatil

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Natural Language Processing, Deep Learning, Computational Storytelling

EDUCATION

Doctor of Philosophy, Computer Science University of Maryland, Baltimore County	GPA: 3.7/4.00	Aug. 2024 – Present <i>Baltimore, MD</i>
Master of Science, Computer Science University of Maryland, Baltimore County	GPA: 3.7/4.00	Aug. 2022 – May 2024 <i>Baltimore, MD</i>
Bachelor of Engineering, Computer Engineering University of Pune, India	GPA: 3.5/4.00	Aug. 2017 – May 2021 <i>Pune, India</i>

RELEVANT COURSES

Machine Learning, Artificial Intelligence, Computer Vision, Data Science, Advanced Algorithms

EXPERIENCE

Graduate Research Assistant University of Maryland, Baltimore County	Jan. 2025 – Present <i>Baltimore, MD</i>
• Work on the Emotional Storytelling project - explore the role of speech prosody, driven by emotional cues, affects the perception of a story. • Work on Dungeons & Dragons (DnD) AI project - develop a reinforcement learning (RL) agent capable of playing DnD, and advancing human-AI collaboration for team based tasks.	
Graduate Teaching Assistant University of Maryland, Baltimore County	Jul. 2024 – Dec. 2024 <i>Baltimore, MD</i>
• Conducted labs and held office hours for the CMSC 202 Computer Science II class. • Ensured adherence to academic integrity standards, identified instances of plagiarism, and reported academic misconduct as necessary.	
Graduate Assistant (Grader) University of Maryland, Baltimore County	Aug. 2023 – May. 2024 <i>Baltimore, MD</i>
• Graded assignments and provided constructive feedback to students for - CMSC 203 Discrete Structures class. • Collaborated with the course instructor to ensure consistency in grading standards and addressed any concerns or questions.	
Graduate Research Assistant University of Maryland, Baltimore County	Jan. 2023 – Aug. 2023 <i>Baltimore, MD</i>
• Implemented machine learning techniques for multi-modal data fusion and utilized visualization tools to study patterns in neuro-biophysical signals, including fNIRS, heart rate (HR), and galvanic skin response (GSR). • Applied statistical methods and tools such as PyWavelets to identify patterns and correlations in data.	
Software Development Engineer Principal Global Services	Jul. 2021 – Jul. 2022 <i>Pune, India</i>
• Developed a custom Salesforce LWC component that imports a CSV file and auto-fills the form, saving users around 2 minutes. • Enhanced the navigation between web pages by 2x by factoring in the availability of a record in the database before the load triggers.	
Research & Development Intern PTC Inc.	Sep. 2020 – Jun. 2021 <i>Pune, India</i>
• Designed and implemented SharePoint solutions to meet the business needs of the company, including architecture design, deployment planning, and system configuration.	

- Collaborated with cross-functional teams, including SMEs, developers, business analysts, and project managers, to ensure accurate and timely documentation of system features and functionalities.

Mathematics Faculty Intern

Jul. 2018 - Feb. 2020

Spectrum Education

Pune, India

- Substituted for faculty in delivering lectures and employed distinctive, student-centered teaching approaches.
- Developed coursework along with the course instructor and organized healthy group discussions.

SELECTED PROJECTS

Shaping Perception of Emotional Storytelling with Synthesized Speech

Feb. 2024 - Present

Advised by: Dr. Lara J. Martin

Examine emotional prosody's (tone, pitch, and rhythm) influence on story perception using a text-to-speech system.

LLM Ablation Study for Generating DnD Game Commands

Sep. 2024 - Dec. 2024

Advised by: Dr. Lara J. Martin

Analyzed the impact of different features such as in-game utterances, player profiles, and game states on predicting the next command in a Dungeons & Dragons (DnD) gameplay setting.

Emotion analysis using EEG based tools

Feb. 2023 - May 2023

Advised by: Dr. Fow-Sen Choa

Built an EEG-based emotion detection system using Support Vector Machines (SVM) and K-Nearest Neighbors (KNN) classifiers for emotion recognition.

OUTREACH AND SERVICE

- Reviewer: Intelligent Narrative Technologies (INT) workshop at AIIDE 2024
- Secondary Reviewer: ACL ARR March 2025, INLG 2024

AWARDS

- Best Poster Award, CSEE Research Day 2025, University of Maryland, Baltimore County, MD